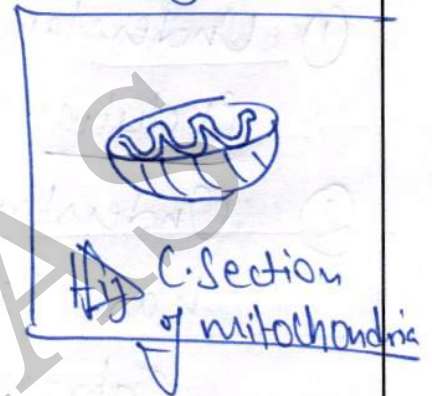


Mitochondrial inheritance

10

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Mitochondria also refers to the "powerhouse of the cell" is a cell organelle present in cell cytoplasm. It contains its own RNA called mitochondrial RNA.



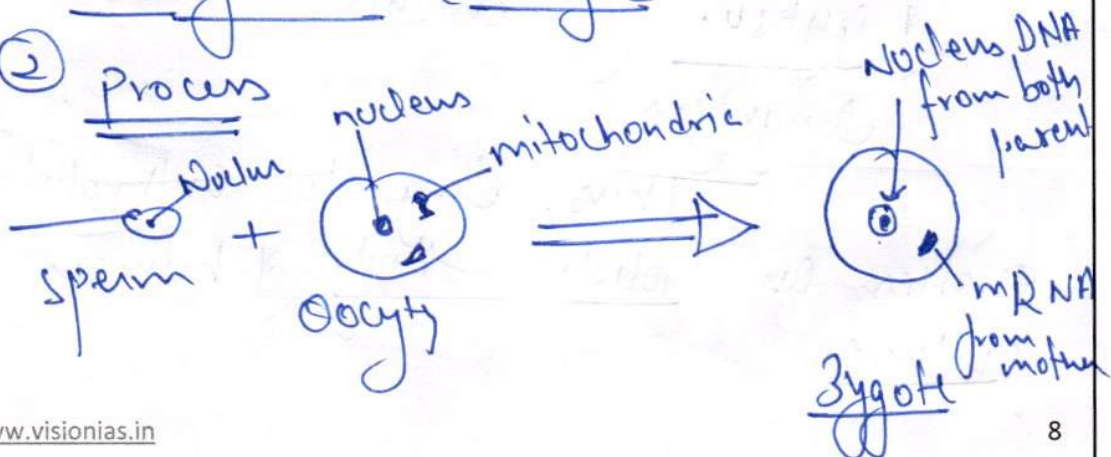
Mitochondrial inheritance

The inheritance pattern of mitochondria is maternal.

Pattern

- ① An individual inherits complete mtDNA from the mother's egg gamete (oocyte).

② Process



Relevance of Studying

① Evolutionary study of human:

the study of mitochondrial Eve highlights single common female ancestor of all humans.

② Three parent baby: techniques like maternal spindle transfer are developed to cure mRNA related diseases.

Thus, the advancements in genetics & mRNA inheritance will enable clearer understanding of human past, present & future.